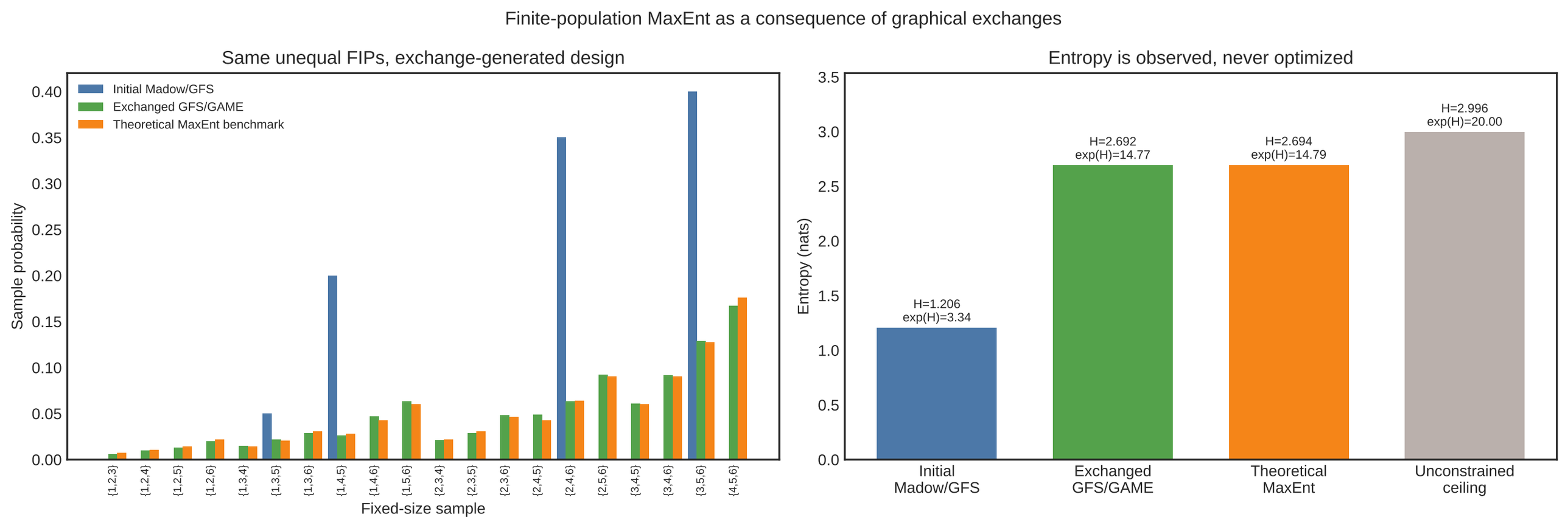
Finite-population MaxEnt as a consequence of graphical exchanges
Same unequal FIPs, exchange-generated design
Initial Madow/GFS
Exchanged GFS/GAME
Theoretical MaxEnt benchmark
Sample probability
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
{1,2,3}
{1,2,4}
{1,2,5}
{1,2,6}
{1,3,4}
{1,3,5}
{1,3,6}
{1,4,5}
{1,4,6}
{1,5,6}
{2,3,4}
{2,3,5}
{2,3,6}
{2,4,5}
{2,4,6}
{2,5,6}
{3,4,5}
{3,4,6}
{3,5,6}
{4,5,6}
Fixed-size sample
Entropy is observed, never optimized
Entropy (nats)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
H=1.206
exp(H)=3.34
H=2.692
exp(H)=14.77
H=2.694
exp(H)=14.79
H=2.996
exp(H)=20.00
Initial
Madow/GFS
Exchanged
GFS/GAME
Theoretical
MaxEnt
Unconstrained
ceiling